A. Winter Sale

time limit per test: 1 second
memory limit per test: 256 megabytes

The winter sale began, and Mrs. Sarah wants to buy a T-shirt for her son. The T-shirt has a ticket that indicates a discount of $X\%$ and its price P after the discount. She now wants to know the price before the discount to know if the discount was worthy. Help her!

Input
Only one line containing two numbers X and P ($1 \le X \le 99, 1 \le P \le 4 \times 10^4$) – the discount percentage, and the price of the T-shirt after the discount.

Output
Print the price of the T-shirt before the discount rounded up to **two decimal places**.

Examples

input
20 80

output
100.00

input
15 20

output
23.53

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    float X;
    long long P;
    long double O, Nm, Dm;

    cin >> X >> P;

    Nm = P*100;
    Dm = 100 - X;

    O = Nm/Dm;

    cout << O;
}
```

let O be Original price,

$P = O - \frac{X * O}{100}$

$O = \frac{P * 100}{100 - X}$

C. Next Alphabet

time limit per test: 1 second
memory limit per test: 256 megabytes

Given a lowercase alphabet character. You have to print the next character in the alphabet.

Input
Only one line containing a lowercase English letter C.

Output
Print the next letter to C in the alphabet.

Example

input
a

output
b

Note
The next letter to **z** is **a**.

#mistake

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    char Ch;

    cin >> Ch;

    Ch = Ch++;
    Ch = Ch % 26;

    cout << Ch;

    return (0);

}
```

ASCII → a = 97 ; issue

Syntax issue

Ch++ ← enough

97 + 25 = 122

We need 122 → 97 (z) (a)

✗ ⇒ d = 100 + 1 → 101 − 97 = 4 → 4 % 26 → 4 → 4 + 97 = 101 → ((ch − a) % 26) + a

z = 122 + 1 → 123 − 97 = 26 → 26 % 26 → 0 → 0 + 97 = 97 → ((Ch − a) % 26) + a

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    char Ch;

    cin >> Ch;

    Ch++;
    Ch = (((Ch-'a') % 26) + 'a');

    cout << Ch;

    return (0);

}
```

→ ((((Ch − 'a') + 1) % 26) + 'a')

6 9 ✓ Syntax ⇒

Q. also possible

H. Data Type Guessing

time limit per test: 1 second
memory limit per test: 256 megabytes

Given three numbers n, k and a. Identify whether the data type of $\frac{n \times k}{a}$ is `int`, `long long` or `double`.

Input

Only one line containing three numbers n, k and a ($1 \leq a, k, n \leq 2147483647$).

absolute limit of int.

Output

Print "`int`", "`long long`" or "`double`" (without quotes) as described in the statement.

Examples

input
3 6 9

output
int

input
100000 200000 4

output
long long

Note

double is when a number has a floating-point.

int Range: $[-2147483648, 2147483647]$.

long long can hold values of a bigger range than that of int.

#Important

Logical Operators

Used to combine multiple conditions together.

- `&&` (Logical AND): True only if **both** sides are true. (`true && false` is false)
- `||` (Logical OR): True if **at least one** side is true. (`true || false` is true)
- `!` (Logical NOT): Reverses the result. (`!true` is false)

int → $(n \times k) \% a = 0$ and ≤ 2147483647

long long → $(n \times k) \% a = 0$

double when none.

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    long long a, k, n, op;
    cin >> n >> k >> a;

    op = n*k;

    ((op) % a == 0)&&((op/a)<=2147483647)&&(cout << "int");
    ((op) % a == 0)&&((op/a)>2147483647)&&(cout << "long long");
    ((op) % a != 0)&&((op!= 0))&&(cout << "double");


    return (0);

}
```